VIRGINIA COMMONWEALTH UNIVERSITY

ASSIGNMENT REPORT

Exploratory Data Analysis of the Iris Dataset

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Course Title	Statistical Modeling and AI-Driven Analytics
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Student Name	Aadithyan Thandayamparambil Suresh Babu
Student ID	2528713
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1. Introduction

The Wholesale Customers dataset contains 440 customer records, two categorical fields (Channel and Region), and six annual spending variables. The analysis asks a simple business question: do Horeca and Retail customers behave differently enough to support separate targeting and merchandising strategies?

2. Data Understanding and Quality Assessment

The raw file had 440 rows and 8 columns; adding readable labels increased the working file to 10 columns without changing the observations. Data quality was strong: there were no missing values, no duplicate rows, and no negative spending entries. Channel was imbalanced (298 Horeca, 142 Retail), and Region was also uneven, with most customers in the 'Other' group. All six spending variables were right-skewed, so medians were more informative than means.

3. Methodology

The work was completed in Python using Pandas, NumPy, Matplotlib, Seaborn, and SciPy. The workflow covered data inspection, descriptive statistics, label creation, median comparisons by channel, Mann-Whitney U tests for channel differences, Kruskal-Wallis tests for region differences, Spearman correlation, and IQR-based outlier screening. Because the distributions were skewed, nonparametric methods were the safest choice.

4. Visualization

The final figure compares median annual spending by channel. Retail customers spend more on Detergents_Paper, Grocery, and Milk, while Horeca customers spend more on Fresh and Frozen. That pattern is the clearest sign that channel segmentation matters more than geography in this dataset.

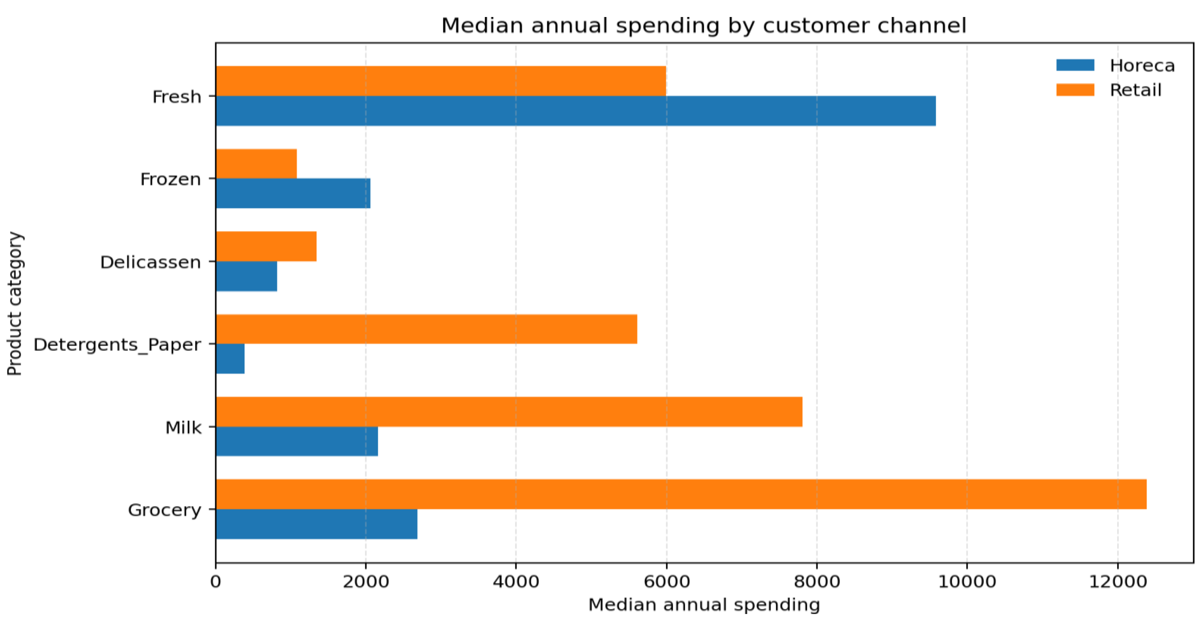

Figure 1. Median annual spending by customer channel.

5. Key Insights

The strongest contrast appears in Detergents_Paper, where Retail's median is 14.56 times Horeca's median. Grocery and Milk also show large Retail premiums, while Horeca spends more on Fresh and Frozen. All six spending categories differ significantly between channels after multiple-testing adjustment, but no category shows a statistically significant regional difference. The strongest correlation is between Grocery and Detergents_Paper (Spearman rho = 0.801), which suggests linked retail purchasing behavior.

6. Conclusions

Overall, the dataset is clean and highly usable for segmentation analysis. Channel is the main driver of spending behavior, while region adds little separation in this sample. A sensible next step would be clustering or predictive modeling on log-transformed spending variables. Generative AI tools were used only for drafting support and formatting assistance; all calculations, checks, interpretation, and final validation were completed manually.